北京交通大學 BEIJING JIAOTONG UNIVERSITY

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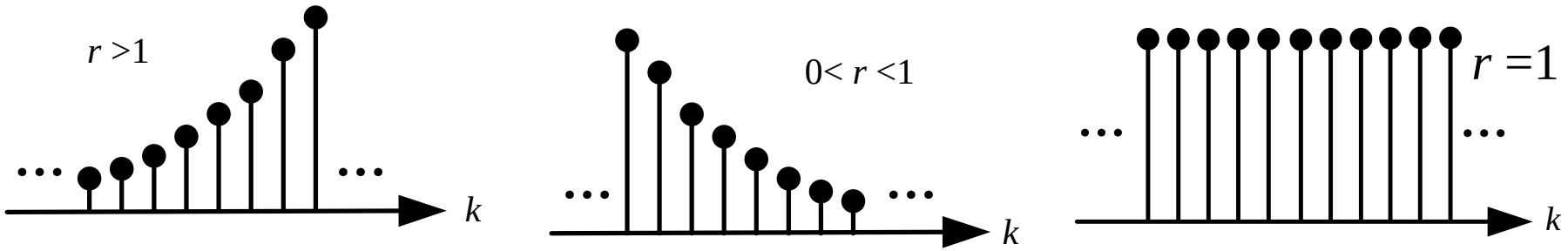
Discrete-Time elementary signals

- Real exponential signals
- Imaginary & sinusoidal signals
- Complex exponential signals
- Unit impulse signal
- Unit step signal
- Rectangular signal
- Ramp signal

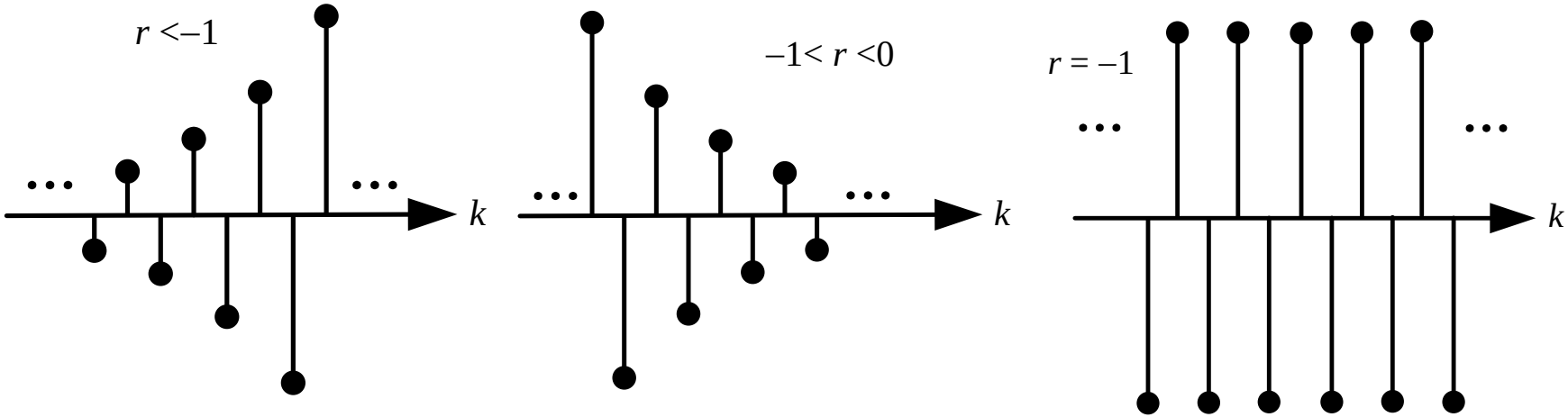
1. Real exponential signals

$$x[k] = Ar^k, \quad k \in Z$$



1. Real exponential signals

$$x[k] = Ar^k, \quad k \in Z$$



Imaginary exponential & Sinusoidal signal

$$x[k] = \mathrm{e}^{\mathrm{j}\Omega_0 k}$$

$$x[k] = A\cos(\Omega_0 k + \varphi)$$

Euler's formula allows you to associate a sinusoidal sequence with a virtual exponential sequence

$$\mathrm{e}^{\mathrm{j}\Omega_0 k} = \cos\Omega_0 k + \mathrm{j}\sin\Omega_0 k$$

$$\begin{cases} \cos\Omega_0 k = \dfrac{1}{2}(\mathrm{e}^{\mathrm{j}\Omega_0 k} + \mathrm{e}^{-\mathrm{j}\Omega_0 k}) \\ \sin\Omega_0 k = \dfrac{1}{2\mathrm{j}}(\mathrm{e}^{\mathrm{j}\Omega_0 k} - \mathrm{e}^{-\mathrm{j}\Omega_0 k}) \end{cases}$$

Imaginary exponential & Sinusoidal signal

D-T imaginary signals can be obtained by sampling.

$$e^{j\omega_0 t}\Big|_{t=kT} = e^{j\omega_0 Tk}, \quad \Omega_0 = \omega_0 T \qquad e^{j\Omega_0 k} = e^{j\omega_0 t}\Big|_{t=kT}$$

periodicity:

If $e^{j\Omega_0(k+N)} = e^{j\Omega_0 k}$, $e^{j\Omega_0 k}$ is a periodic signal。

$e^{jW_0 N} = 1$, $\Omega_0 N = m2\pi$ (m is an integer), it is a periodic signal.

If $\Omega_0/2\pi = m/N$, this discrete signal is a periodic sequence;

When N、m is an irreducible integer, the period of the signal is N

Imaginary exponential & Sinusoidal signal

$e^{j\omega_0 t}$ It must be a periodic signal

If the frequency ω_0 is different, the signal is different

The rhythm of signal oscillation increases with the increase of frequency ω_0

If $\Omega_0/2\pi = m/N$
This discrete signal is a periodic sequence

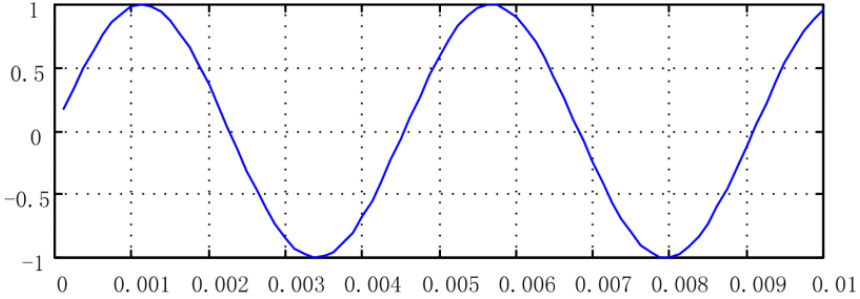
$e^{j\Omega_0 k}$ Not necessarily a periodic signal

The imaginary exponential sequence of frequency Ω_0 is the same as the exponential sequence of frequency $\Omega_0 \pm 2n\pi$

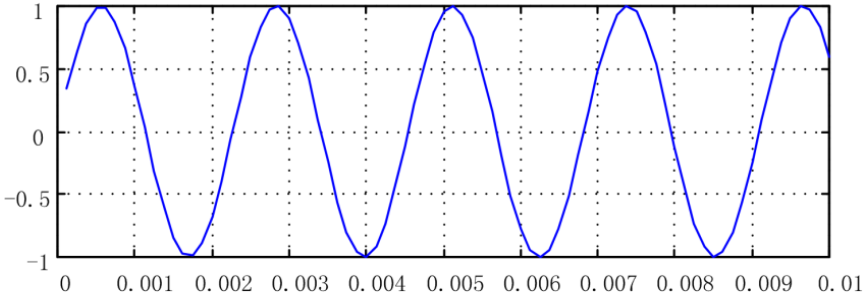
The sequence oscillation rhythm does not necessarily increase with the increase of frequency Ω_0

Imaginary exponential & Sinusoidal signal

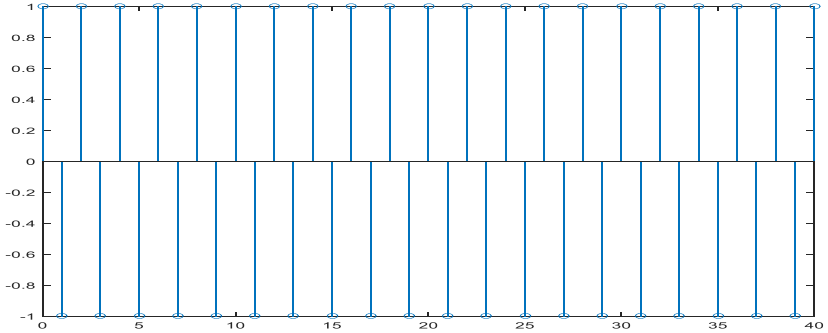
$$x_1(t) = \mathbf{sin}(2\pi \cdot 220t)$$



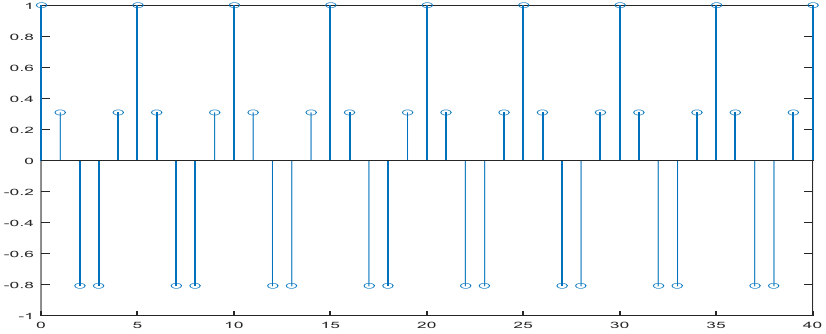
$$x_2(t) = \mathbf{sin}(2\pi \cdot 440t)$$



$$x_1[k] = \cos(\pi k)$$



$$x_1[k] = \cos(1.6\pi k)$$



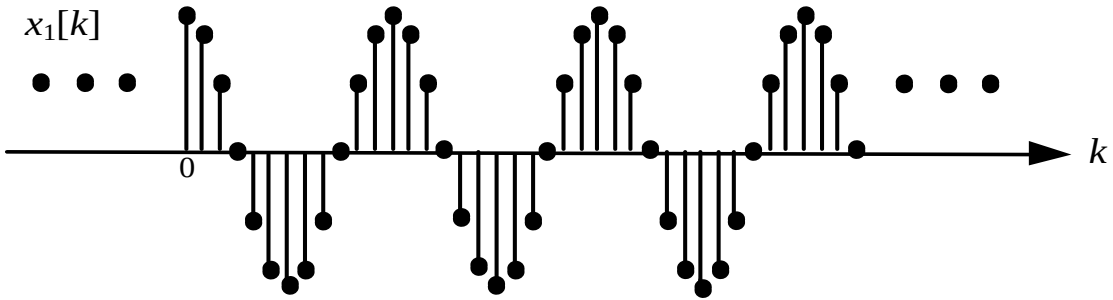
Determine whether the following discrete sequence is a periodic signal

(1) $x_1[k]=\cos(k\pi/6)$

$\Omega_0=\pi/6$

$W_0/2p = 1/12$, It is a rational number，

So the period of discrete sequences $x_1[k]$ is $N=12$。



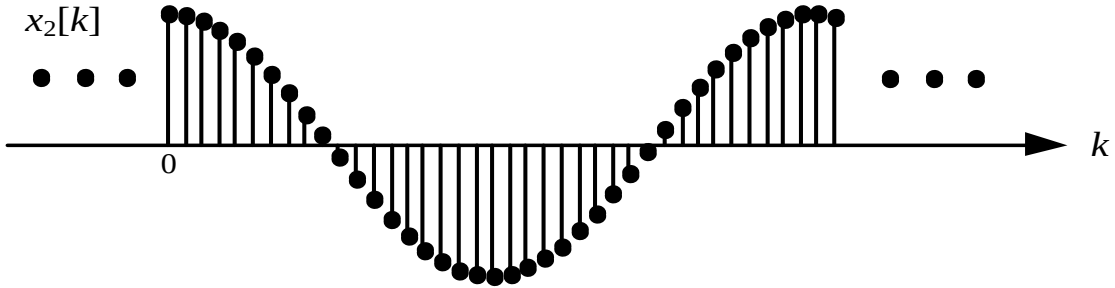
Determine whether the following discrete sequence is a periodic signal

(2) $x_2[k]=\cos(k/6)$

$\Omega_0 = 1/6$

$W_0/2p = 1/12p$, 1/12p is a rational number,

So discrete sequences $x_2[k]$ is a non-periodic sequence。



Determine whether the following discrete sequence is a periodic signal

It can be obtained by using the same method as above

(1) Sampling cos(πt) at T=0.08s to obtain a sequence $x_1[k]$

$$\frac{\Omega_0}{2\pi}=\frac{0.08\pi}{2\pi}=\frac{1}{25}=\frac{m}{N}$$ $m=1$ The period is $N=25$

(2) Sequence obtained by sampling cos(πt) at T=0.16s $x_2[k]$

$$\frac{\Omega_0}{2\pi}=\frac{0.16\pi}{2\pi}=\frac{2}{25}=\frac{m}{N}$$ $m=2$ The period is $N=25$

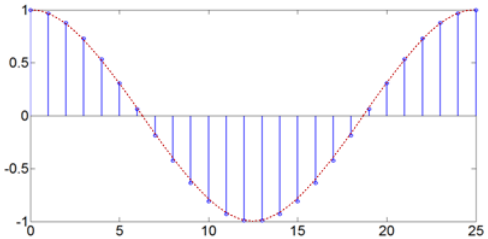
(3) Sampling cos(πt) at T= 0.24s to obtain a sequence $x_3[k]$

$$\frac{\Omega_0}{2\pi}=\frac{0.24\pi}{2\pi}=\frac{3}{25}=\frac{m}{N}$$ $m=3$ The period is $N=25$

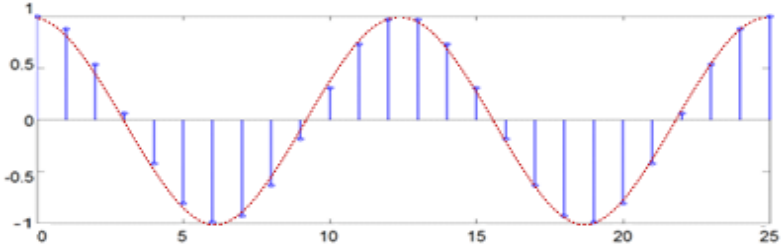
What does the value of m mean?

Discrete signals

$$\frac{m}{N} = \frac{1}{25}$$



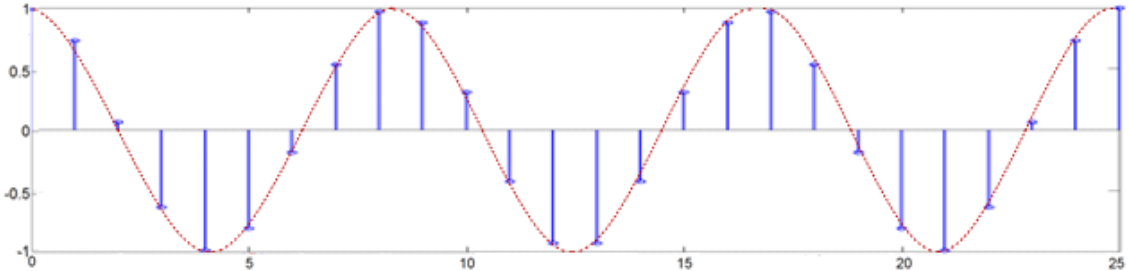
$$x_1[k] = \cos(\pi t)\big|_{t=0.08k} = \cos(0.08\pi k)$$



$$x_2[k] = \cos(\pi t)\big|_{t=0.16k} = \cos(0.16\pi k)$$

$$\frac{m}{N} = \frac{2}{25}$$

$$\frac{m}{N} = \frac{3}{25}$$



$$x_3[k] = \cos(\pi t)\big|_{t=0.24k} = \cos(0.24\pi k)$$

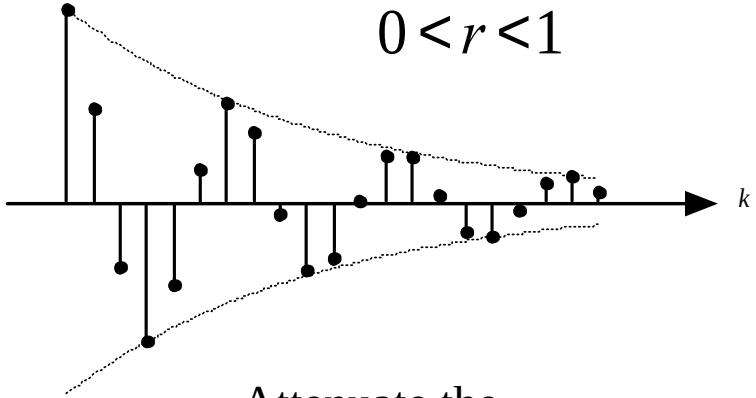
m indicates the number of cycles in a discrete sequence N that contains the original continuous period signal

Complex exponential Signals

$$x[k] = Ar^k e^{j\Omega_0 k} = Az^k, \quad z = re^{j\Omega_0}$$

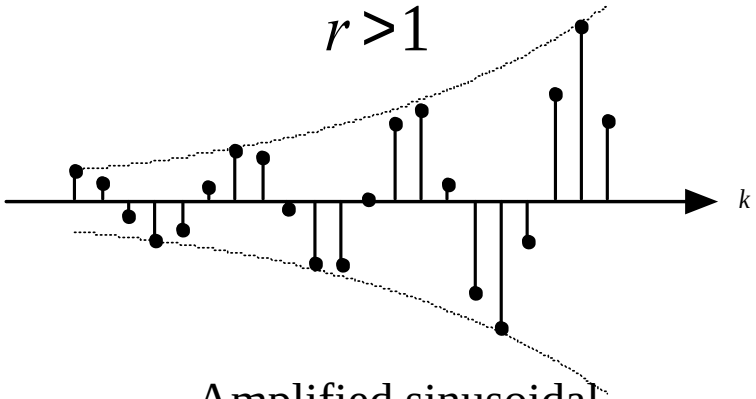
$$Ar^k e^{j\Omega_0 k} = Ar^k \cos(\Omega_0 k) + jAr^k \sin(\Omega_0 k)$$

$0 < r < 1$



Attenuate the sinusoidal signal

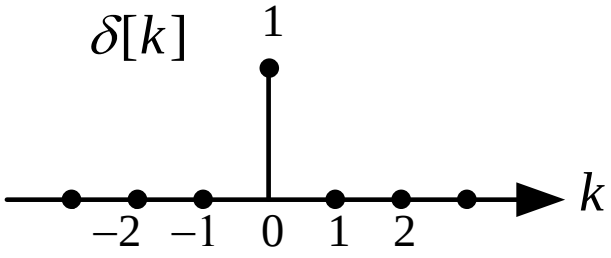
$r > 1$



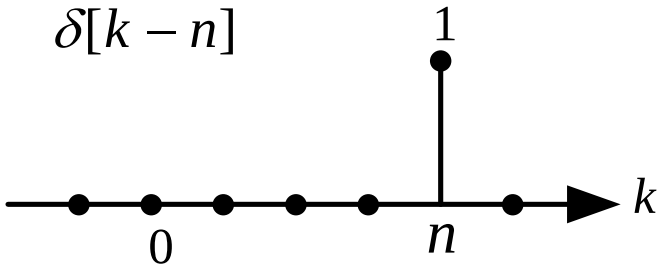
Amplified sinusoidal signal

D-T Unit impulse signal

$$\delta[k]=\begin{cases}1 & k=0\\ 0 & k\neq 0\end{cases}$$

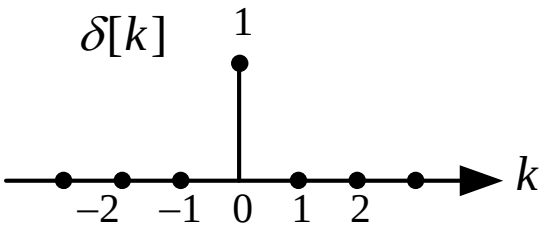


$$\delta[k-n]=\begin{cases}1 & k=n\\ 0 & k\neq n\end{cases}$$



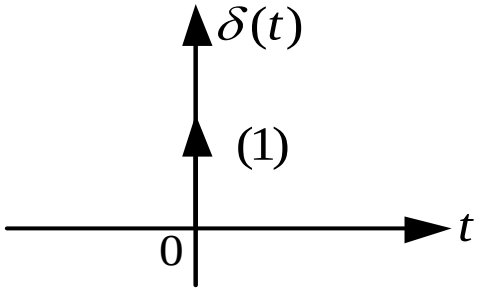
Differences between $\delta(t)$ and $\delta[k]$

$$\delta[k]=\begin{cases}0 & k\neq 0\\ 1 & k=0\end{cases}$$



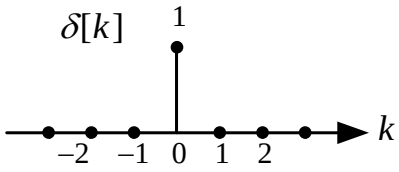
$$\delta(t)=0\ ,\quad t\neq 0$$

$$\int_{-\infty}^{+\infty}\delta(t)\,\mathrm{d}t=1$$

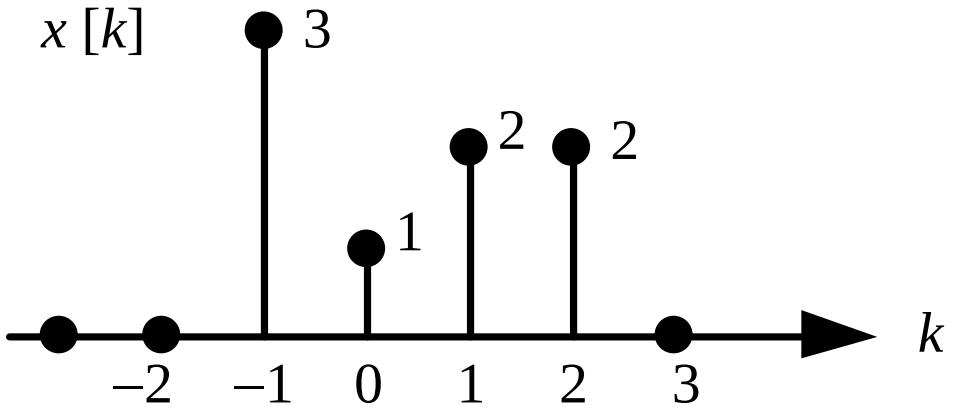


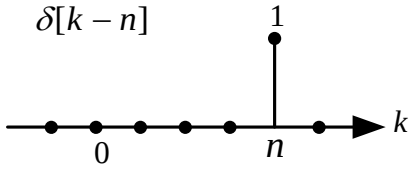
D-T Unit impulse signal

- Function: Represents an arbitrary discrete time series



D-T unit pulse signal



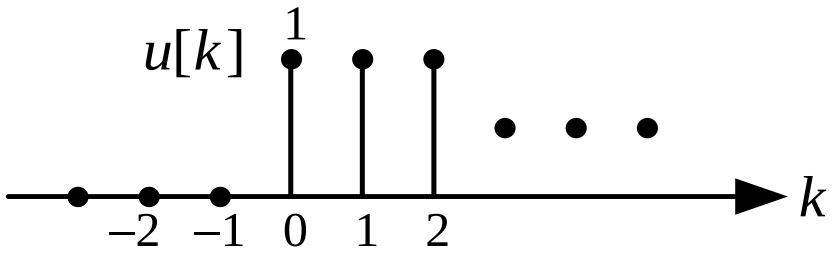


D-T unit impulse with a delay.

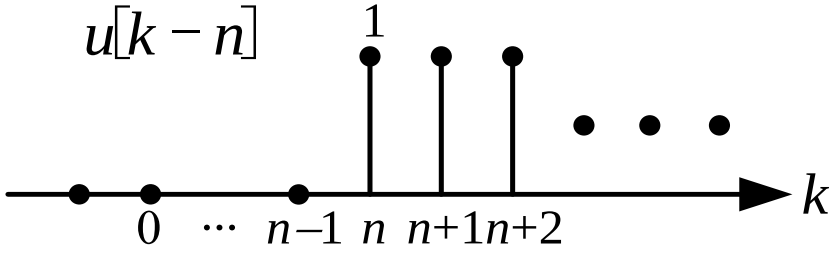
$$x[k] = 3\delta[k+1] + \delta[k] + 2\delta[k-1] + 2\delta[k-2]$$

D-T unit step signal

$$u[k]=\begin{cases}1 & k\geq 0\\ 0 & k<0\end{cases}$$

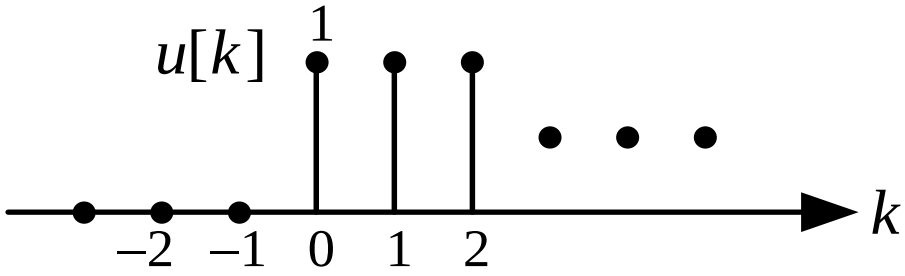


$$u[k-n]=\begin{cases}1 & k\geq n\\ 0 & k<n\end{cases}$$

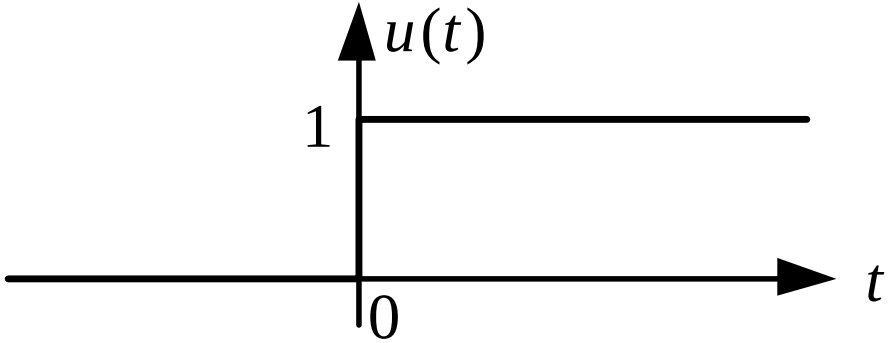


D-T unit step signal

$$u[k]=\begin{cases}1 & k\geq 0\\ 0 & k<0\end{cases}$$



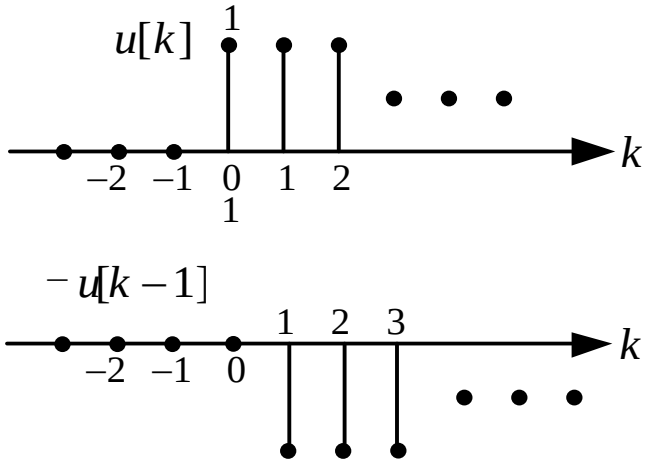
$$u(t)=\begin{cases}1 & t>0\\ 0 & t<0\end{cases}$$



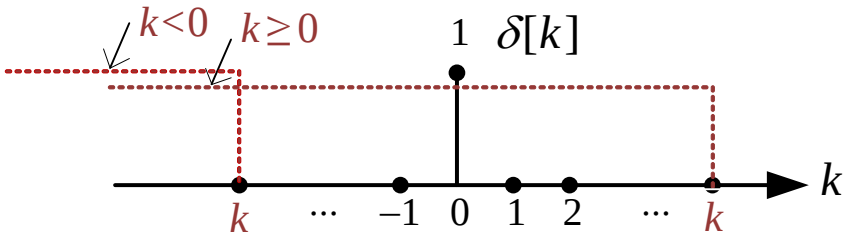
D-T unit step signal

➢ $\delta[k]$与$u[k]$的关系

$$\delta[k]=u[k]-u[k-1]$$



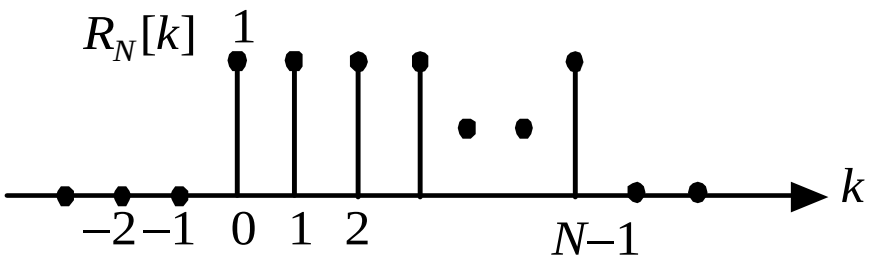
$$u[k]=\sum_{n=-\infty}^{k}\delta[n]$$



D-T rectangular signal

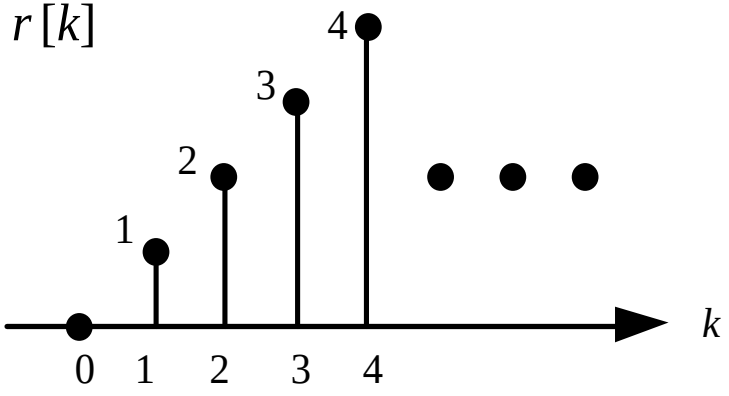
$$R_N[k]=\begin{cases}1 & 0\le k\le N-1\\ 0 & \text{otherwise}\end{cases}$$

$$R_N[k]=u[k]-u[k-N]=\sum_{m=0}^{N-1}\delta[k-m]$$

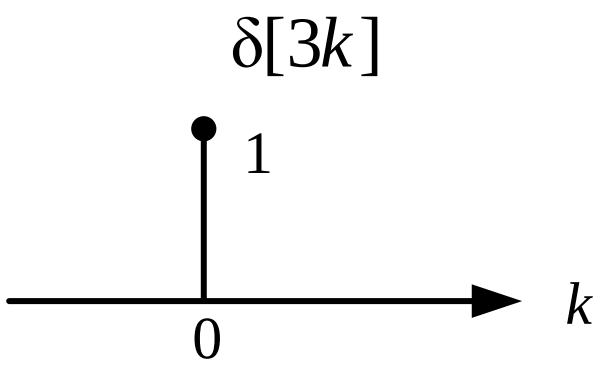


D-T ramp signal

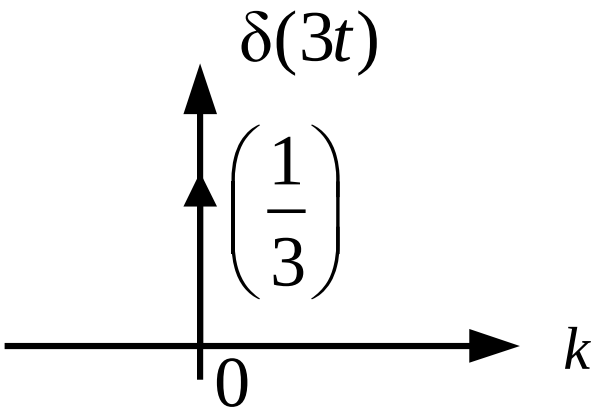
$$r[k] = k \cdot u[k]$$



Example: Graph the following discrete time series



$\delta[3k]$



$\delta(3t)$

Thanks

Some of the materials cited in this course are the teaching accumulation of the main lecturer for many years, which come from a variety of media and the exchanges of colleagues, peers and friends, and it is difficult to indicate the source one by one, so I hereby explain and thank you!